

Workforce Development Analysis: Salaries & Tuition Assistance in Montgomery County

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DATA 205

CRN: 33334

1. Introduction and Project Overview

Workplace equity remains critical across public institutions, especially in local government employment. This report examines pay equity and tuition assistance programs in Montgomery County through the analysis of two datasets: employee salaries (2024) and tuition assistance records (2013-2024). The study addresses critical questions about gender pay disparities and equitable access to professional development opportunities. The primary goals were to:

- Detecting and quantifying gender-based salary disparities.
- Examine how tuition assistance funds are distributed across departments.
- Explore relationships between departmental pay levels and educational benefit utilization.

These goals are important for informing equitable policy decisions, enhancing transparency, and ensuring fair access to professional development opportunities in Montgomery County. The project uses R with the tidyverse and ggplot2 for statistical analysis and visualization, and Tableau for interactive dashboards. OpenRefine was also used for data optimization and cleaning.

2. Data Sources

This analysis uses two publicly accessible datasets from the Montgomery County Open Data Portal:

Tuition Assistance (2013–2025 (updated))

- 4,470 records of approved tuition reimbursements
- Key variables: Department, Major, Degree, School, and Cost
- Last updated: February 2025
- Source: https://data.montgomerycountymd.gov/Education/Tuition-Assistance/p7z5-tjrz/about_data

Employee Salaries (2024)

- Complete 2024 compensation records for Montgomery County employees
- Key variables: Department, base salary, gender
- Source: https://data.montgomerycountymd.gov/Human-Resources/Employee-Salaries-2024/2nq6-auk8/about_data

Key Cleaning Steps

1. Department Standardization
 - Consolidated department name variants into 25 standardized names across both datasets to unify them for analysis using OpenRefine
 - Used the Department variable in both datasets as the common key to allow for accurate merging and comparison
2. Salary Data Processing
 - Converted salary strings to numeric format

3. Initial EDA through dashboards in Tableau

Top Funding Allocations:

1. Police Department: \$1,179,935 total funding
2. Fire/Rescue Services: \$716,914
3. Health & Human Services: \$634,261

Degree-Level Patterns:

Degree Type	Courses Funded	Total Cost	Avg. Cost/Course
Master's	945	\$1,085,831	\$1,149
Bachelor's	1,442	\$1,049,634	\$728
Associate's	680	\$293,026	\$431

Top Funded Majors:

1. Business Administration/Management
2. Criminal Justice

3. No Major Specified

Highest Median Salaries (Base Pay):

1. Legislative Oversight: \$149,888 (No overtime)
2. Management & Budget: \$148,985 (No overtime)
3. Technology Services: \$138,580 (Avg. \$456 overtime)

Overtime Leaders:

Department	Median Overtime
Correction & Rehabilitation	\$24,259
Board of Elections	\$21,235
Fire/Rescue Services	\$21,213

Gender Disparities (Selected Departments):

Department	Female Salary	Male Salary
Legislative Oversight	\$139,196	\$184,640
County Executive	\$122,962	\$170,888
Police	\$78,357	\$105,166

4. Primary Analysis

The statistical analysis revealed some significant gender-based pay disparities across Montgomery County departments:

Police Department: 26,808.33 mean difference (Male:\$105,166 vs Female:\$78,357)

Highly significant ($t = -14.70$, $p < 0.001$)

Large effect size (Cohen's $d = 0.79$)

Female officers would need 2-3 additional promotions to match male starting salaries

Lifetime earnings gap: ~\$536,160 over a 20-year career

Legislative Oversight: 45,444 (Male: \$ 184,640 vs Female: \$ 139,196)

Marginally significant ($p = 0.067$) due to small sample size (6 males, 10 females)

Very large effect size (Cohen's $d = 1.37$)

Power analysis shows 63% power - requires 6+ additional male records for 80% power

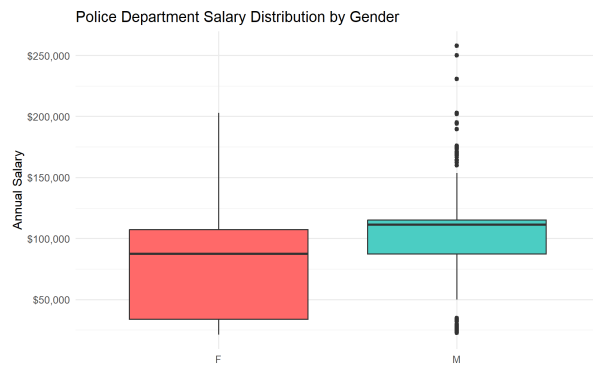


Figure 1: Boxplot visualization of gender pay gap in the Police Department

Key findings from our analysis:

Funding Distribution:

- The Police Department received the most funding (\$1,179,935)
- Master's degrees accounted for the highest total cost (\$1,085,831) but had lower enrollment than Bachelor's degrees

Salary-Tuition Relationship:

- No significant correlation ($r = -0.09$, $p = 0.663$)

5. Final Analysis and Policy Implications

Systemic Patterns Identified:

- Structural Pay Gaps: Gender disparities are shown clearly in the police department and legislative oversight, but to validate legislative oversight, we need a power of at least 80%, but we only have 63% currently (small group size)
- Benefit Equity: Tuition access appears equitable regardless of salary

Departmental Variations:

- Police show the largest absolute gap
- Legislative Oversight shows the largest percentage gap

Recommended Actions:

Immediate (0-6 Months):

1. Police Department Equity Audit:

- A comprehensive audit of police salaries should control for rank, years of service, and overtime hours. Success will be measured by reducing the gender pay gap's effect size from Cohen's $d^* = 0.79$ (large) to $d < 0.5$ (moderate) and narrowing the 95% confidence interval to $\pm \$5,000$.

2. Legislative Oversight Data Collection:

- Given the small sample size (10 women, 4 men), priority should be given to collecting 6 additional male salary records to achieve 80% statistical power. This will clarify whether the observed \$45,444 gap (women earn 75.4% of male salaries) reflects systemic disparity or sampling limitations.

Long-Term (12+ Months):

1. Automated Equity Dashboard

Develop an R Shiny dashboard to monitor:

- Real-time pay gap metrics
- Tuition program participation
- Promotion pipeline demographics

2. Longitudinal Study:

- Propose tracking employee cohorts to assess how tuition assistance impacts career trajectories over time.

6. Conclusion

This comprehensive analysis confirms statistically significant gender pay disparities in Montgomery County government, with particular findings in the Police Department. While tuition assistance programs demonstrate equitable access across salary levels, the compensation disparities reveal systemic inequities requiring immediate intervention.

Key conclusions:

1. Police Department results are both statistically and practically significant, demanding urgent action
2. Legislative Oversight findings, while limited by sample size, suggest potentially severe disparities needing verification
3. Tuition benefits are distributed equitably, but could be optimized for impact

The power analysis provides crucial context; we can be highly confident in the Police Department findings, while the Legislative Oversight results (63% power) require cautious interpretation and additional data collection.

These evidence-based recommendations position Montgomery County to:

- Address verified disparities in policing
- Investigate potential gaps in other departments
- Maintain equitable professional development access

By implementing these measures, Montgomery County can establish itself as a leader in public sector pay equity while fostering employee growth through its tuition assistance program. Future research should incorporate longitudinal analysis and additional control variables to strengthen these findings.

References

Montgomery County Government. *Employee Salaries - 2024*. Montgomerycountymd.gov, 24 Feb. 2025, https://data.montgomerycountymd.gov/Human-Resources/Employee-Salaries-2024/2nq6-auk8/about_data. Accessed 11 May 2025.

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